# Does the learner know how often he solves a puzzle?

The coaching app's solve chances, replayed over the player's puzzle record

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**Abstract**

The coaching app sets the difficulty of each puzzle set from a Glicko-2 rating: it aims where its learner says the player will solve 75% of the time. Replayed over the profile as the app holds it (100 puzzle attempts on seven days, logged as he played), the rating serving reads gave him a mean solve chance of 58.0% on puzzles he solved 96.0% of. It sits close to the difficulty he was served, so the sets aim far below him. The learner's view of his themed sets, which it already keeps, gave 82.4%, still low, with about half the log loss. The test committed before the run used a record rebuilt from the app's memory cards, which turned out to keep almost only his solves, because a missed puzzle comes back for another review; its verdicts do not stand. A second round, fixed before any of its attempts exists, tests the view on the attempts the app records next. The app's owner adopted the view before it, on the profile's look; round 2 decides whether it stays.

## 1 The question

The app's learner keeps a Glicko-2 rating for each skill and turns it into the chance that the player solves an item of a given difficulty: Glicko's expected score against the item's rating, with the item's own deviation. Serving reads that chance backwards. A bucket's puzzle set aims at the difficulty he should solve 75% of the time (85% while the bucket is new), and draws the unseen puzzles nearest to it. The profile's puzzle totals count 441 solved of 467.

The learner keeps two estimates of each skill. The *headline rating* folds every channel and counts a result in an announced set at the Lichess discount, so that a record earned in announced sets does not overstate how he plays when nothing is announced: in a bucket's own set a solve moves it a fifth as far as one in an unannounced set and a miss 0.7 as far, and for an obvious theme such as mate in N a tenth and 0.4. The legacy attempt log cannot say whether an attempt was a first try, so each of its attempts counts at half of that. Beside it, a *view* per channel folds only that channel's first exposures, each at full weight. Serving reads the headline. The learner's own notes expect the discount to read a player served at 75% in themed sets about 60 points under his strength. When the learner gives a solve chance, how often does he solve? And is the rating serving reads the right one?

## 2 The record and the rule

**The record.** The rule was committed before any solve chance was computed. It named a record rebuilt so that every first try has its time. The app keeps a memory card for every puzzle he has been served, with its last review's time and its passes and fails, and the bank gives each puzzle's

bucket and Lichess rating. A card reviewed exactly once holds one first try with its time and result; a card reviewed more than once keeps only its last review's time, so it was left out. That leaves 365 of the 429 cards: 24 with the app's own attempt event and 341 rebuilt as the event the app writes today, 365 first tries from 2026-07-21 to 2026-10-05. The cards, like everything else round 1 reads, come from the profile at commit `d8b7712`, the one round 1 measured; what the app records after it is round 2's.

**The instrument.** For every first try, in time order, the app's own learner at commit `4ba253b` is folded on the record before it, at the moment of the attempt, and gives two solve chances for the puzzle: from the headline rating and from the view of the attempt's channel (for a bucket's own set, the themed view). An attempt is scored when its skill already has a rated attempt in that channel, so both exist. A solve scores $-\ln p$ and a miss $-\ln(1-p)$, as the learner's own self-check scores them.

**The candidate.** Read a set's solve chance, and so its target, from the view of the set's own channel, falling back to the headline where the view has no attempt yet. It fits nothing.

**The rule.** The days with scored attempts alternate between two halves. (1) The headline under-predicts if the share solved minus its mean chance has a 95% interval over 1,000 bootstrap draws of the days (seed 7) entirely above zero, over-predicts if entirely below, and is calibrated within the noise otherwise. (2) The view replaces the headline only if, on each half, its mean log loss is below the headline's with the interval of the difference entirely below zero. Named beside the verdicts, deciding nothing: the same replay on the profile as the app holds it, scores against a running rate (the share of the skill's earlier attempts he solved), a reliability table, the buckets, a best-fitting shift, the learner's self-check, and where each estimate would aim a set today.

## 3 Round 1, and why its verdicts do not stand

On the rebuilt record (354 scored attempts on 21 days) he solved 99.7%, and the headline's mean chance was 52.0%: a gap of 47.7 points (40.5 to 56.8), so by rule 1 the headline under-predicts. By rule 2 the view passes on both halves: its log loss minus the headline's is $-0.426$ on the first (11 days, $-0.569$ to $-0.266$) and $-0.509$ on the second (10 days, $-0.729$ to $-0.348$).

Neither verdict stands. A missed puzzle becomes a redemption card that comes back until he solves it, while a solved one retires, so a card reviewed once is nearly always a solve (Table 1). Leaving out the cards reviewed more than once removed 25 of the 26 puzzles he ever missed, which is the outcome both rules measure. The rebuilt record's 99.7% is that selection, and on a record with almost no misses a higher chance can only score better, so rule 2 could hardly fail. The memory a scheduler keeps is shaped by the results it reacts to; a record of first tries has to come from a log written as they were played.

| memory cards | cards | misses |
|---|---|---|
| reviewed once (round 1's record) | 365 | 1 |
| reviewed more than once (left out) | 64 | 26 on 25 cards |

Table 1: Where the misses sit on the memory cards. A miss returns as a redemption card, so the cards reviewed once hold almost none.

## 4 The profile as the app holds it

The profile also holds what the app logged as he played: the legacy attempt log (106 entries, 94 of them on cards reviewed once and 12 on the others, all agreeing with the cards in time, bucket, rating and result), the puzzle totals from before the log, which the learner folds as a seed, and the attempt events. Every attempt in it was logged when it was played, so nothing in it selects on the result. Replayed the same way, it gives 100 scored attempts on seven days, from 2026-08-28 to 2026-10-05, with four misses: he solved 96.0%.

The headline's mean chance was 58.0%, a gap of 38.0 points (31.3 to 42.7); the view's was 82.4%, a gap of 13.6 (8.9 to 22.1). Both under-predict, the headline by far more (Table 2). The view's log loss is about half the headline's, a difference of −0.292 (−0.366 to −0.161), and the reliability table shows where: the headline put most of his attempts below a 70% chance, and he solved nearly all of them (Table 3). The same holds in every bucket with more than one attempt (Table 4). The view scores only about as well as the running rate, though: the share of the skill's earlier attempts he solved.

| estimate | mean chance | Brier | log loss |
|---|---|---|---|
| headline rating (serving reads it) | 58.0% | 0.191 | 0.568 |
| view of the themed sets | 82.4% | 0.069 | 0.276 |
| running rate | 93.1% | 0.064 | 0.298 |
| share solved | 96.0% | | |

Table 2: Each estimate's mean solve chance and scores on the profile's 100 scored attempts. Lower is better for both scores.

| | headline rating | | | view | | |
|---|---|---|---|---|---|---|
| solve chance | attempts | mean chance | solved | attempts | mean chance | solved |
| 0.0 to 0.5 | 22 | 45.2% | 90.9% | 3 | 37.4% | 100.0% |
| 0.5 to 0.6 | 37 | 55.2% | 100.0% | 0 | – | – |
| 0.6 to 0.7 | 29 | 63.7% | 93.1% | 2 | 65.5% | 100.0% |
| 0.7 to 0.8 | 8 | 74.2% | 100.0% | 17 | 75.8% | 100.0% |
| 0.8 to 0.9 | 4 | 81.9% | 100.0% | 67 | 85.0% | 94.0% |
| 0.9 to 1.0 | 0 | – | – | 11 | 92.1% | 100.0% |

Table 3: The scored attempts binned by each estimate's solve chance.

Why the headline reads him so low shows in its own fold. It starts each skill at the median rating of the puzzles he was served, and a solve in a mate set that the log cannot call a first try moves it a twentieth of a full step. So it follows the puzzles up and never gets ahead of them: on the profile it

| bucket | attempts | mean rating | solved | headline's chance | view's chance |
|---|---|---|---|---|---|
| fork | 11 | 906 | 100.0% | 60.0% | 84.0% |
| hanging piece | 16 | 1511 | 87.5% | 59.4% | 82.4% |
| mate in one | 37 | 1436 | 94.6% | 49.4% | 82.9% |
| mate in two | 11 | 693 | 100.0% | 61.4% | 85.8% |
| mate in three | 1 | 1075 | 100.0% | 47.6% | 27.0% |
| planning | 24 | 716 | 100.0% | 68.5% | 81.5% |

Table 4: Each bucket's scored attempts, the mean Lichess rating of its puzzles, the share solved and each estimate's mean chance.

sat on average 60 points above the puzzles he was asked about (their mean rating 1131), where Glicko gives a little over half, while the view sat 287 points above. Figure 1 shows it on mate in one, where he solved 35 of 37. The shift that best fits the headline's chances, taken off every puzzle's rating, is 500 or more (385 to 500, the grid's edge); the view's is 305 (160 to 500, the grid's edge).

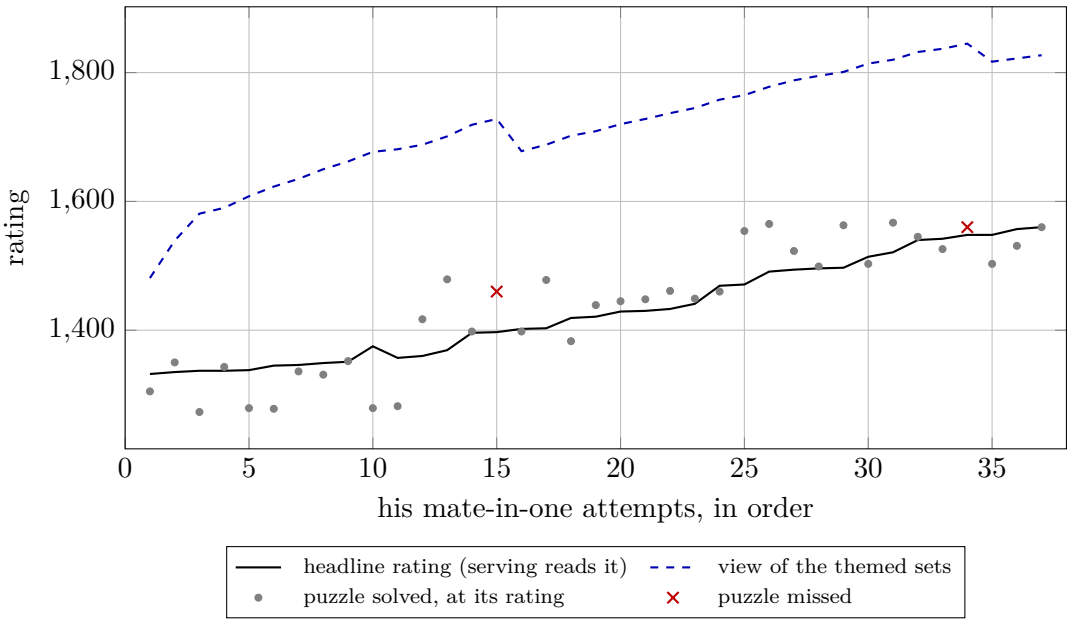
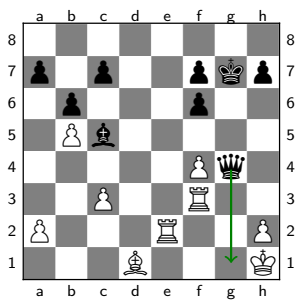



Figure 1: The two estimates before each mate-in-one attempt on the profile, and the puzzle's Lichess rating. The headline follows the puzzles; the view sits well above them.

The learner's own self-check, which folds each channel's earlier attempts and predicts the later ones, says the same of the views: on its 40 held-out attempts it predicted 85.3% and he solved 92.5%, and its Brier score, 0.075, is no better than its constant's, 0.070. Figure 2 shows the solved puzzle the headline gave the lowest chance.



**The puzzle:** mate in one, rated 1479 on Lichess; Black to move.

**The solution:** 30. . . ♕g1# (green), `g4g1`.

**Before it:** the headline rated him 1369 and gave 35%; the view rated him 1701 and gave 78%. He solved it.

**FEN:** `8/p1p2pkp/1p3p2/1Pb5/5Pq1/2P2R2/P3R2P/3B3K b - - 0 30`

Figure 2: The solved puzzle on the profile the headline gave the lowest chance, as he faced it, after the opponent's move.

## 5 What it means for the app

The sets aimed far below him. Today the view aims a set 18 to 287 points higher than the headline in the six buckets that have a themed view (Table 5). The other five have no attempt in their own sets since the log began, and the app aims them from their stored totals read alone at full weight, 86 to 309 points higher than the headline. The bank can serve every one of those targets. The view still under-predicts him, so sets aimed from it should still land above 75%, but nearer.

By its rule this round adopts nothing: its test did not stand, and the profile's look, named beside it, decides nothing. On that look the app's owner chose to adopt the view at once: since chess PR 78 the app aims a bucket's set from its themed view, and the skill map's next difficulty reads the same estimate; since PR 79, a bucket with no themed view aims from its stored totals instead of the headline. Commit `4ba253b`, which this paper measures, has both. Round 2 decides whether the view stays; it scores no attempt in a bucket without one. It takes the attempts the app records after the last one in the profile at commit `d8b7712` (played on 2026-10-07, Los Angeles time), each folded on everything before it as the app holds it, and runs once they fall on at least ten days. Rules 1 and 2 apply unchanged: sets now aim harder, and each estimate is still scored on the puzzles he was served. A third rule, fixed after round 2's first attempts were recorded and before any of their results was read, tests whether the chance holds for one-move puzzles and longer ones alike, which a target per depth would need; the project's README states it.

| | headline | | themed view | | stored totals | | |
|---|---|---|---|---|---|---|---|
| bucket | rating | aims at | rating | aims at | rating | aims at | hardest in the bank |
| planning | 918 ± 222 | 725 | 1123 ± 94 | 930 | – | – | 1878 |
| hanging piece | 1608 ± 245 | 1415 | 1797 ± 108 | 1604 | – | – | 1899 |
| defensive move | 1096 ± 246 | 903 | none | – | 1182 ± 70 | 989 | 1898 |
| mate in one | 1587 ± 247 | 1394 | 1833 ± 82 | 1640 | – | – | 1896 |
| fork | 1006 ± 280 | 813 | 1282 ± 119 | 1089 | – | – | 1897 |
| mate in two | 796 ± 290 | 603 | 1083 ± 118 | 890 | – | – | 1897 |
| mate in three | 1066 ± 314 | 873 | 1084 ± 212 | 891 | – | – | 1899 |
| pawn endgame | 1223 ± 318 | 1030 | none | – | 1418 ± 112 | 1225 | 1887 |
| discovered attack | 1056 ± 328 | 863 | none | – | 1362 ± 112 | 1169 | 1899 |
| London System | 1056 ± 328 | 863 | none | – | 1362 ± 112 | 1169 | 1494 |
| f2 and f7 | 1040 ± 337 | 847 | none | – | 1349 ± 85 | 1156 | 1889 |

Table 5: Each puzzle bucket as the profile round 1 reads stood when it was synced, on 2026-10-13 (Los Angeles time): each estimate's rating with its deviation, the difficulty a set would aim at from it (the core target, clamped to what the bucket can serve), and the hardest puzzle the bank holds. The stored totals, read alone, are shown where the app aims from them: in a bucket with no themed view.

# 6 Limitations

- One player, served puzzles far below him: the profile holds four misses, so its log losses rest on a handful of them.
- The profile's attempts fall on seven days, so the bootstrap over days has few to draw from and its intervals are rough.
- The log cannot say which attempts were retries; the replay counts them as the app does, at half weight in the headline and in full in the view.
- A puzzle's published Lichess rating stands for its difficulty, as in the app. An item-level model would need more than one attempt per puzzle.
- Round 1's record was rebuilt from memory that selects on the result. Round 2 reads only the app's own log, written as the attempts are played.